LAB 3 — COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Self-Service Coffee Kiosk System

Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is divided into clear layers: the Touchscreen UI handles customer interaction, the Order Manager handles business logic and order coordination, the Payment Service handles credit-card payment processing, and the Menu & Pricing Database and Receipt Printer provide data and hardware services.

Two Reasons for the Choice

1. Clear separation of concerns: Customer interaction, order processing, payment, data access, and receipt output are separated into components. This makes the kiosk easier to understand, test, maintain, and modify.

2. Suitable for a small centralized kiosk: The system has a limited and well-defined workflow with one kiosk interface, one payment mechanism, one menu data store, and one receipt printer. Layering provides modularity without the operational complexity of microservices.

Security Advantage

Security-sensitive payment processing is isolated in the Payment Service. The Order Manager requests payment through a defined interface without directly handling the internal payment implementation. This concentrates validation and secure payment handling at the payment boundary.

Performance Benefit

Most interactions remain local to the kiosk, avoiding repeated network communication and service-discovery overhead associated with microservices. Menu information can also be retrieved efficiently from the local data component.

Component & Interface Summary

Component	Responsibility
Touchscreen UI	Customer menu/order interaction.
Order Manager	Order validation and coordination.
Payment Service	Credit-card payment processing.
Menu & Pricing DB	Coffee types, sizes and prices.
Receipt Printer	Receipt output hardware.

Interfaces: IOrderSelection (UI → Order Manager), IMenuQuery (Order Manager → DB), IPayment (Order Manager → Payment Service), IReceipt (Order Manager → Receipt Printer).